

A Prostate Cancer Game Based on Multiple Decision-Making and Planning Skills

Maria Andréia Formico Rodrigues
PPGIA (CCT)
Universidade de Fortaleza
Fortaleza-CE, Brazil
andreia.formico@gmail.com

Samuel Pessoa Madeira
Ciência da Computação (CCT)
Universidade de Fortaleza
Fortaleza-CE, Brazil
spmadeira@unifor.br

Juliana Amaral de Figueiredo
Arquitetura e Urbanismo (CCT)
Universidade de Fortaleza
Fortaleza-CE, Brazil
jup.amaral77@gmail.com

Isabel Veras Beleza
Medicina (CCS)
Universidade de Fortaleza
Fortaleza-CE, Brazil
isabel_beleza@hotmail.com

Alice Vasconcelos Rolim
Ciência da Computação (CCT)
Universidade de Fortaleza
Fortaleza-CE, Brazil
alicevrolim@edu.unifor.br

Marcel Costa Lima Leite Juca
Ciência da Computação (CCT)
Universidade de Fortaleza
Fortaleza-CE, Brazil
marceljuca@edu.unifor

Abstract—The scientific literature on games and health applications aimed at helping decision-making for prostate cancer is relatively modest, therefore, still requiring more significant contribution and exploration. This work presents a prostate cancer game based on decision-making and planning skills designed to replicate classic real-world prostate symptoms, treatment patterns, and care pathways. Players on the role of medical professionals have to manage the patients' health and cope with the care of multiple prostate cancer patients concurrently. Which patient to help first and which medical tests to request to decide the diagnosis and correct medical action are tasks that the player has to manage. Since every decision has its consequences, the ultimate goal is to make decisions without losing patients' lives.

Index Terms—prostate cancer, interactive game, decision-making, choices & consequences, planning skills

I. INTRODUCTION

Prostate cancer is one of the most common cancer types among men [1]. More specifically, the second and fourth most commonly occurring, respectively, in men and overall. About one man in eight will be diagnosed with prostate cancer during his lifetime. It develops when abnormal cells multiply and accumulate to form a tumor which can be cancerous or benign. In more detail, prostate cancer affects the prostate gland, located in the lower part of the male abdomen, just below the bladder, and in front of the rectum [2].

Worldwide, about 75% of prostate cancer cases occur from the age of 65. Some of these tumors can proliferate, spread to other organs, and even lead to death. Most tumors evolve slowly (it takes about 15 years to reach 1cm³), not becoming an immediate threat to man's health. Anyway, early diagnosis is essential to maximize the likelihood of cure, and the harm that this disease can cause to the individual's health should not be neglected [2]. Recently, according to the American Cancer Society, the number of men who will die from prostate cancer will hit a record high over the last two decades [3].

In parallel with that, computer games can create active learning platforms applied to the health area [4]. Instead of a didactic presentation that requires memorization or assimilation of facts out of context, games promote situated learning in which players can discover and learn through exploration and experimentation. When playing a game, players can cause desirable and undesirable health consequences through decision-making without putting actual patients in danger. In addition, games can help players make meaningful connections between real-life and simulated events, thereby increasing the likelihood that knowledge and skills experienced in the game world are retained and applied in the real world.

Some authors discuss the benefits and the importance of developing more games aimed at the health care area, mainly in Oncology [5], [6], [7], [8], [9]. Those authors present several reasons explaining why computer games may provide benefits, for example: for doctors, by helping to improve the management of patient care; for medical institutions, by reducing treatment costs and increasing the satisfaction of patients and health professionals; for the Ministry of Health, by health promotion through education; for society, by increasing the level of participation in social activities through the learning of social skills; and for patients, by promoting continuity of care without the time and place restrictions.

The scientific literature on games and health applications aimed at helping the decision-making for prostate cancer is relatively modest [10], thus, still requiring greater contribution and exploration. This work presents *CancerSP* (Fig. 1), a prototype game based on decision-making designed to replicate classic real-world prostate symptoms, treatment patterns, and care pathways. Players on the role of medical professionals (Urologists, Oncologists, and Medical students) have to manage the patients' health and cope with the care of multiple prostate cancer patients concurrently. Which patient to help first and which medical tests to request to decide the diagnosis and correct medical action are tasks that the player

has to manage in this game. Since every decision has its consequences, the ultimate goal is to make decisions without losing patients' lives. Having a game experience for decision-making within the prostate cancer context may help medical professionals and students practice prioritization in choosing the right actions in the correct order and time, aiding in making the best decisions for patients.

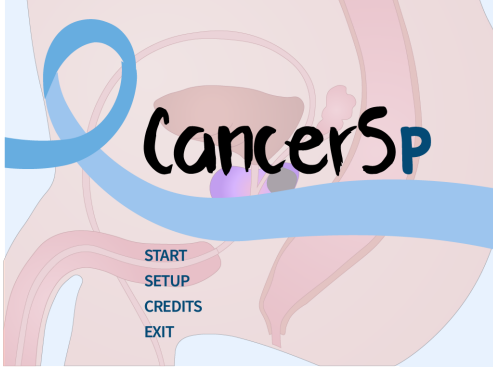


Fig. 1. Start screen of *CancerSp* game.

II. RELATED WORK

This section presents related work focused on *serious games* and *mobile phone applications* developed to increase prostate cancer awareness.

A. Serious Games

From simulated experiences on the common impacts on patients' quality of life with prostate cancer, the *Time After Time* game [6] focuses on guiding players to choose the most accepted cancer treatment strategy, considering the various side effects it may cause. The side effects addressed in the game relate to the following cancer treatments: radical prostatectomy, brachytherapy, external radiation therapy, and watchful waiting. The game shows cards with examples of side effects so that the user can classify them on a 5-point scale (in which one means "without problems" and five means "with major problems"). The first round ends when the player finishes sorting the side effect cards for all treatments and periods. In the end, the player can view the result of the game, which includes the cancer treatments with the side effects that best meet the user's preferences (in a way, empowering the player) and a list of questions created for the doctor containing doubts about the side effects found in the cards.

The *Prostate Cancer Evaluation and Education* game [7] has been designed in collaboration with its target audience towards increasing awareness of prostate cancer among Afro-Caribbean men and encouraging symptomatic men to seek medical help. In practice, this game features eight levels. In seven of these levels, the user is presented with an informational area on how to play and with directed questions about the assessment of the risk of prostate cancer, patient concerns, family history, digital rectal examination (DRE), urinary complaints, and prostate-specific antigen levels (PSA).

The last game level is used to evaluate the user's responses and encourages the patient to do an active search for medical follow-up.

B. Mobile Phone Applications

The authors of a systematic review [10] of mobile phone apps for quality of life and well-being assessment, especially in prostate cancer patients, argue that people with cancer tend to be more open to strategies that positively influence their therapeutic process. Moreover, they suggest that game developer should take special care in playtesting with this audience and implement approaches more patient-centered.

WebChoice is a digital tool that has been developed in cooperation between users and healthcare professionals [11]. It contains tabs for recording symptoms and problems, sending messages, writing a personal diary, opening public forums, accessing bibliographies on each health issue, and oncology nurses and doctors counseling. Studies conducted using this tool have shown that men with prostate cancer have low levels of social support and high levels of suffering due to the symptoms of this disease, suggesting that these users may find such types of tools particularly useful.

Interaktor is an application for smartphones and tablets developed to support the management symptoms of patients with various types of cancer, including prostate cancer [12]. Concepts of participative and individual-centered care inspired the development of the intervention. It includes four generic components: (1) Self-assessment, (2) A risk assessment model, (3) Evidence-based self-care advice & links to relevant web pages, and (4) Graphs showing the history of symptoms reported to patients and healthcare professionals. Authors report that the application promoted the patient's participation in their care through continuous and convenient contact with health professionals.

A slightly different approach to assess the usability and patient experience is presented in [13], with a mHealth application named *myPROHealth*. This application focuses on men who are about to start androgen deprivation therapy, *i.e.*, a therapy that increases the risk of adverse metabolic effects among patients with prostate cancer. The application's features allow patients to enter information about their physical activity and recent meals and monitor their treatment evolution. Although the number of users who participated in the user tests was minimal (only five patients, with an average age of 62 years, and five doctors), the results from the qualitative assessments revealed that patients, in general, valued the holistic care provided by the application.

These works use different initiatives, such as guiding users to choose the most accepted strategy of prostate cancer treatment, supporting the management symptoms of patients with cancer, among others. However, the remaining gap is the lack of previous prostate cancer games based on decision-making and planning skills. This work fills a gap by designing a prostate cancer game that focuses on decision-making, and classic real-world prostate symptoms, treatment patterns, and care pathways. Finally, while management games let players

manage every aspect of a person's life, this work aims to mix decision-making with managing multiple patients with potential prostate cancer concurrently. From managing time with various patients to handling complicated patient care situations, players can expect to experience problems related to what action they should do first using planning skills. In our game, patients' symptoms and family medical history provide the initial information for recognizing what medical actions must come first to identify priority clinical case scenarios for making better treatment decisions.

III. GENERAL DESCRIPTION

Prostate cancer is usually diagnosed by prostate biopsy prompted by a blood test to measure PSA and/or DRE [14]. PSA is a protein produced by normal, as well as malignant, cells of the prostate gland. DRE involves a doctor feeling the prostate through the wall of the back passage (rectum) to verify if one has a prostate problem or prostate cancer.

Symptoms fundamentally depend on the location and extent of the disease. Most cases are diagnosed by screening and, thus, patients, in general, have no suspicious symptoms of the disease. In case of a possible localized cancer disease, the most common symptoms originate from the lower urinary tract, showing dysuria (pain when urinating), increased urination, hesitation, and reduced urinary jet, erectile dysfunction, etc. [2]. Cases of locally advanced disease can present hematuria (blood in the urine), urinary retention, pelvic pain, and hematospermia (blood in the sperm). On the other hand, in cases of advanced disease, the patient's symptomatology will depend on the location of the metastasis.

Most prostate cancers are diagnosed after a PSA elevated for age and/or prostatic nodule detected in DRE, being performed transrectal prostate biopsy by ultrasound. Patients with metastatic disease have significantly elevated PSA and can obtain histological evidence for diagnosis by biopsy or site of metastatic disease. After confirming the diagnosis of prostate cancer, an evaluation should be carried out through laboratory tests and imaging exams to stage the disease and plan treatment. The imaging tests to be ordered will depend on the clinical presentation of each patient and each risk of cancer stratification [14].

The *Gleason score* [15] is a score given to prostate cancer based on its microscopic appearance. It is important because higher scores are associated with a worse prognosis. In terms of local staging, magnetic resonance imaging has often been used since it allows a better correlation with the volume of the cancer disease. In addition, another potential advantages are the staging of the primary tumor, which can assist in cancer treatment in some cases.

To fight against prostate cancer and, consequently, save lives, through the exercise of clinical diagnosis, decision making, and cancer treatment, the player must manage the following resources available in the Game home interface:

- Medical Tests, Imaging and Interventions:** PSA (or Blood Test) & DRE (or Touch Test), Ultrasound (US) & Magnetic Resonance Imaging (MRI), and Biopsy.
- Medical Treatments:** Based on the results obtained from PSA, DRE, US, MRI, and Biopsy, one can apply one or more treatments appropriate to the prostate cancer type, such as Radiotherapy, Surgery, Chemotherapy, and Hormone therapy.

Clinical Reasoning is an important aspect of health professional education and practice [16]. The basics and fundamentals, when put together, reflect a physician's ability to make decisions and diagnoses. Depending on the results obtained from the medical tests, imaging, interventions, and patient's age, family medical history, and symptoms, a decision has to be made.

IV. ART & DESIGN

The design of this game was conceived by one of the authors of this work. The game art was developed as a collection of image sprites and icons, most of them created by the authors, and some others owned by third parties offered through the Website [17], [18]. The light blue ribbon as a symbol of prostate cancer awareness was our inspiration to choose the color palette and create the art of our game as an emblem of the prostate cause.

In our game, the player takes on the role of saving patients with suspected prostate cancer. The music sets the tone for the importance of medical care [19]. So, we opted for creating a Start screen still, with a near-dead silent and a simple menu (Fig. 1), which includes the following options: *Start*, *Setup*, *Credits*, and *Exit*. After starting the game, the melody bounces according to the player's initiatives in saving patients' lives or when losing lives [20]. The musical theme sets the tone for what is to come during gameplay.

The decisions of images and sounds joined on the game screens were made considering how both aspects continually influenced each other, mainly focused on the decision-making and consequences context, when trying to save the lives of patients with different types of prostate symptoms concurrently. The sound also provides feedback and alerts for the player, including button-press clicks, signs, etc.

In this prototype version of the game, we modeled 18 classic clinical case scenarios in the context of prostate cancer to put player's diagnostic, management, and treatment skills to the test. We describe these case scenarios as follows: healthy prostate, prostate with painful condition (with different levels of pain), prostate bleeding (with the small, medium, and large bloody flow), prostate with varying sizes of tumor and stages of development, prostate with inflammation, and prostate with different types of metastasis, as shown in Fig.2.

In the upper panel of the Game home screen, there are different clinical cases of patients presented simultaneously (in Fig. 3, Fred, Peter, and Sam), being three at a time. That is, multiple clinical case scenarios are dynamically spawned on the Game home screen. The player must manage the patients' health as quickly and precisely as possible. Every decision has its consequences, so the ultimate goal is to make decisions without losing the patients' lives.

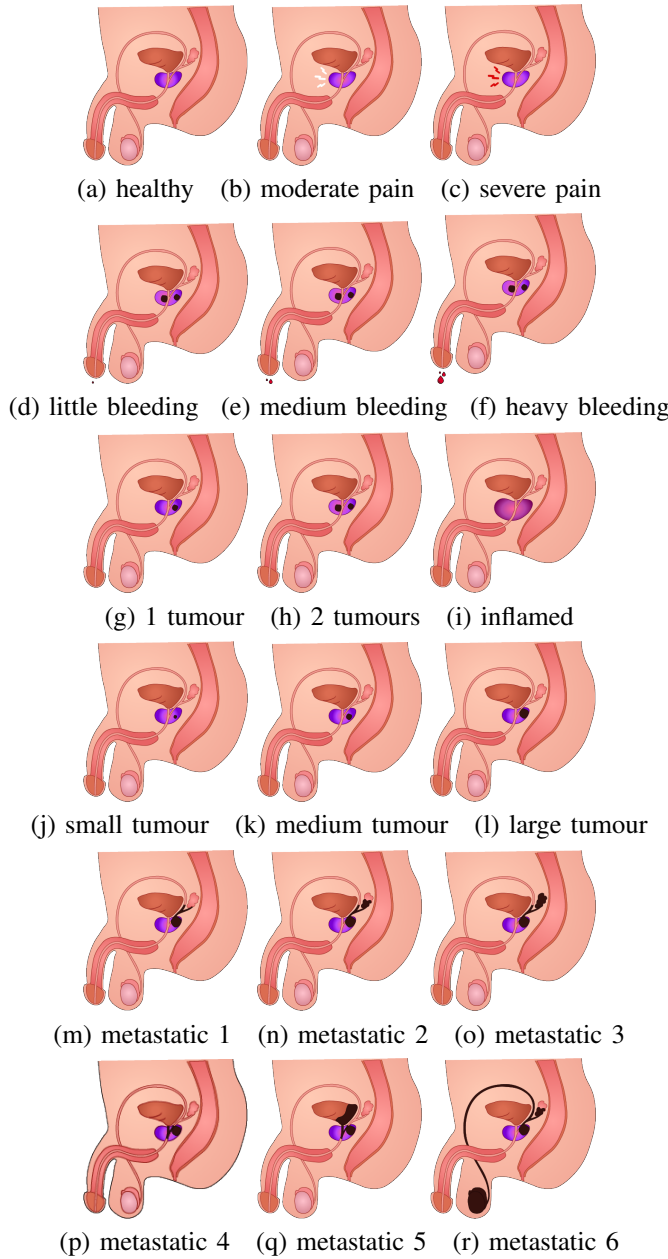


Fig. 2. Classic clinical case scenarios of prostate cancer, designed for *CancerSP* game.

To help in the decision-making process, the initial data input for each patient is presented to the player on the top of the Game home panel, in each patient's lane, as follows: *Name* (names are fictional), *Age*, *Symptoms*, and *Family Medical History*. The player can scroll the panel down to facilitate the reading and monitoring of all clinical cases.

Icons are an essential part of game user interfaces, visually expressing objects and actions. We use icons on the six buttons displayed in the lower area of the Game home screen (Fig. 3) to control game mechanics. More specifically, Figs. 4.a, b, c, d, e, and f detail the game features available to the player for clinical diagnosis, intervention, and treatment:

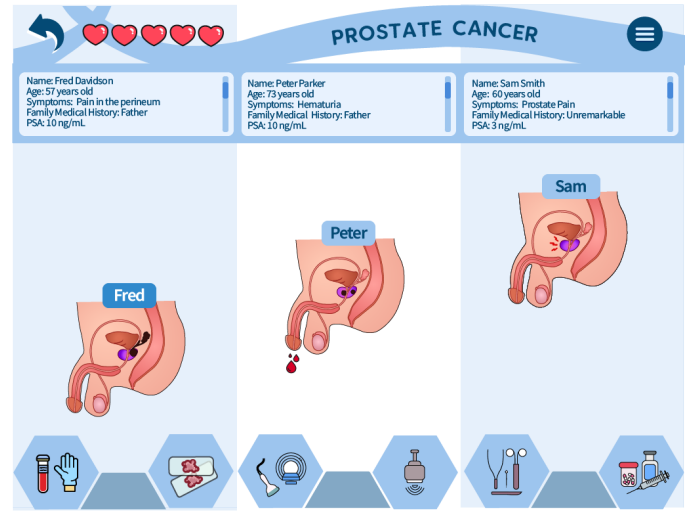


Fig. 3. Game home screen of *CancerSP* game.

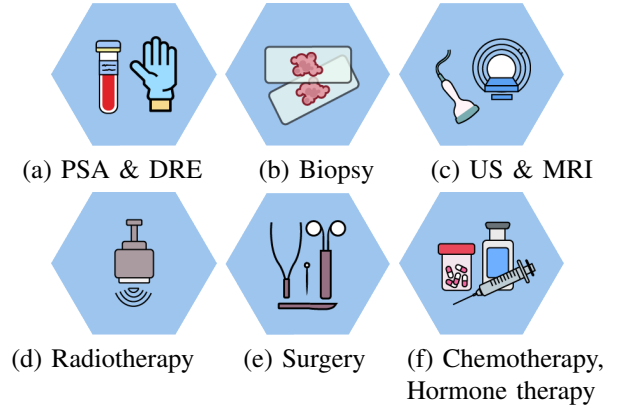


Fig. 4. Medical tests and imaging, biopsy, and treatments for prostate cancer, available in the *CancerSP* game.

Medical tests (PSA & DRE) and imaging (US & MRI), intervention (Biopsy), and treatments (Radiotherapy, Surgery, Chemotherapy, and Hormone therapy). As the player interacts with the game environment using these action buttons, updated information about the clinical case scenarios is displayed on each patient's lane.

There is a *return* icon on the top left side of the Game home screen to make it possible to go back to the Start screen. On the right side of the *return* icon, five heart icons show how many life points the player begins the game with. In other words, how many times can the player make wrong decisions to the end of losing all the patient's lives before being eliminated from play (game over). There is a *menu* icon on the top right side of the Game home screen with the following options: *Help*, *Restart*, and *Exit*.

V. GAME DEVELOPMENT

CancerSP is a computer game made using *Unity* [21], currently still the most popular platform to build games with.

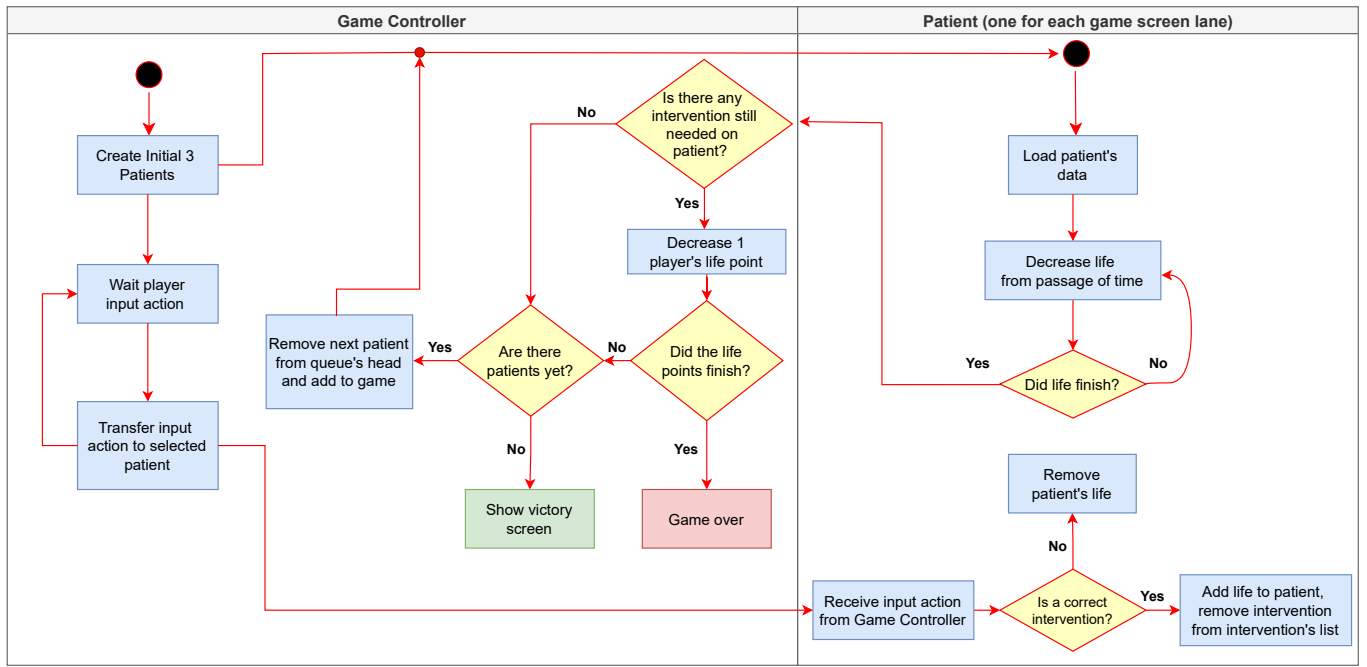


Fig. 5. General flowchart of the *CancerSP* game.

With multiple patients to care concurrently (each one in a different lane), the player needs to be more detached to activate and manage multiple game elements simultaneously. The diagram shown in Fig. 5 is a general flowchart representing the flow of the prostate cancer game from start to finish. According to the player's actions, each patient's health status can improve or worsen during gameplay. Thus, to select and take care of any patient, the player must first click on a particular dynamic patient's sprite (Fig. 3) to activate the diagnosis and treatment options. Currently, the clinical cases are hardwired in-game logic. The static clinical cases contain the patient information and the order of operations necessary to consider the patient as "cured".

Illustrative Example: The following patient's clinical case scenario is shown in the middle lane of Fig. 3: a 73-year-old man named Peter Parker reported having hematuria, with a family medical history of prostate cancer (father). His past medical history was unremarkable. To investigate this clinical case, the player must first select the patient by clicking on his name (Peter) on the blue label at the top of the patient's sprite. A similar procedure must be done to interact with the other two patient cases, selecting any of their names: Fred or Sam. By pressing the PSA & DRE button icon (Fig. 4.a), a PSA level 10 ng/mL is shown in the top panel of the Game home screen. And, when scrolling down (the player can scroll down the panel content wherever necessary), the DRE revealed a slightly enlarged and significant asymmetry prostate. By asking for a prostatic biopsy (Fig. 4.b), the player is notified that the biopsy result was positive for malignancy in both lobes, clinical stage T2, and revealed a *Gleason score* of 7 (4 + 3) adenocarcinoma. Based on this information, the

player can make a treatment decision, that is, for this case report, to undergo surgery (radical prostatectomy) (Fig. 4.e), followed by Hormone therapy (Fig. 4.f).

Basically, from the player perspective, the patient's choice criteria will most likely be based on the severity of his health status (the closer the patient's sprite is to the end of the lane, the higher the risk of patient's health, requiring priority emergency care). As time goes by, the game dynamics are as follows: all the "patients" will fall vertically in an animation movement of translation. When the patient reaches the base of the Game home screen, there are two options: either the player will save the patient's life or lose it. In the latter case, at the top of the Game home screen, one life point (*heart* icon) will be removed for each life lost.

If the player makes a wrong decision, this causes the player to lose one of the five life points. As a result, one life point is taken away from the Game home screen. The game continues, and the player starts again with all the remaining lives. Likewise, if the player makes a wrong medical decision, this decreases the patient's life and may cost another life point. The game continues like this until there are no more life points available (*i.e.*, the player lost five patients' lives) or until the player completes the game (*i.e.*, the player saved many patients' lives and lost less than five life points). If the player succeeds in completing the gameplay, a victory is achieved, as shown in Fig. 6. Otherwise, a message is shown regretting the loss of patients' lives (Figs. 7 and 8).

Other essential elements in the game appear when the game is over. The game ends when the clinical case scenarios presented in-game are over, and the player successfully managed to save the majority of patient's lives, or when five life points



Fig. 6. Game over of *CancerSP* game: the player saved many lives.



Fig. 7. Funeral scene, displayed every time a life is lost, followed by a minute's silence.

(i.e., the lives of five patients) are lost. By its end, the player who has not lost their patients' lives celebrates life with the audience, as shown in Fig. 6. On the other hand, the player who has lost patients' lives must observe a minute's silence to show respect for each of those who have died (Fig. 7). To engage players, we tailored a customized message ("You were not able to save *Patient's Name*"), that varies according to the name of the lost patient ("Fred", in the case of Fig. 7). In addition, when the game is over, the screen shown in Fig. 8 will appear. This final screen was designed to reflect the whole



Fig. 8. Game over screen of *CancerSP* game: player lost many lives.

process of decision making and treatment to be given and the meaning of death since there's no more significant loss than the loss of life.

A. Animations

Most of the animations are used when the patient's state changes. In groups of three, sprite images representing different clinical case scenarios of patients with prostate symptoms are moved down vertically, representing their respective life journeys. The player must be aware that the health of the three patients needs to be managed simultaneously. We designed this type of game interaction to create a more engaging experience by motivating players to get better and faster at making decisions, similar to actual Hospital-Based Emergency Care.

In patients with prostate cancer, the disease can evolve or involute, depending on the player's actions. Given the clinical scenario of each patient, every time the player makes the correct decision, the patient gains lifetime (his sprite goes up, instead of falling down the lane). On the other hand, when the player makes a wrong decision, the patient loses his lifetime (his sprite falls faster). When a patient is near the end of his lane, we scale his sprite down so that he can cross the quadrilateral object located between the action buttons and leave the Game home screen, representing a gate metaphorically to the life or death.

To represent these changes in the patients' clinical condition and provide the player with a more engaging and interactive experience, we designed and implemented different animations focused on cancer symptoms using sprite animations. The sprite animations were created using keyframe animation in *Unity*. *Unity* interpolates the data from one keyframe to the next to play the animation when the animation plays. Among the collection of sprites (Fig. 2) we have designed for the animations, the classic prostate symptoms implemented in this version of the game using animations were the following:

- **Prostate with pain:** Represented by a throbbing visual effect. By improving this symptom, this visual effect is turned off;
- **Prostate with hematuria:** Represented by three sprites with incremental intensities of bleeding. By improving this symptom, the same animation approach is used, however, in the reverse order of keyframes;
- **Prostates with tumors that increase in size, number, and/or undergo metastasis over time:** Represented by sprites with three incremental levels of tumor size, number of tumors, and types of metastasis.

Finally, through *Events* in *Unity*, visual effects start and stop spawning particles periodically using a time delay, as shown in Fig. 6 in the form of light blue awareness ribbons scattered around the phrase "Congratulations! Many lives were saved!".

VI. PLAYTESTING

A. Participants

First, for having an initial measure of the perceived usability of the game, we have conducted a playtesting session with only one specialist user who detected the need to calibrate the

movement speed of the patients' sprites to generate a better game experience.

Initially, the time limit for saving each patient was the 30s. This value proved to be unsuitable even for players who like to play at a fast pace. It has generated a very high level of difficulty and, more importantly, it failed to convey the patient's input data adequately and clearly. Thus, the time limit was adjusted to 60s, which proved to be much more compatible with the game's goals without harming the sprite animations and player interaction quality. It is important to reinforce that, in addition to the challenge of controlling multiple patients' sprites at a time, there is also the challenge of evaluating each clinical case scenario and be confident in making the appropriate decision during gameplay by reading and processing each medical information displayed on the Game home screen.

Since a qualitative study with five users is likely to uncover 85% of the usability problems in a design [22], following that, we conducted playtesting with a small sample of five users (3 males and two females). Overall, participants showed a similar level of knowledge about prostate cancer. The goal was to determine the main issues of our game prototype and fix them.

B. Questionnaires

First, we began with the application of an online questionnaire using Google Meet to allow audio and visual communication between each participant and the facilitator, with the following qualitative questions:

- 1) Did you like the game's visual aesthetic?
- 2) Were the sound effects and musical theme appropriate for the game?
- 3) Did you find useful the learning design of the game?
- 4) Did you find the prostate cancer cases representative?
- 5) Did you like the experience?
- 6) Would you have suggestions for game refinement and improving?

Then, we applied an online version of the SUS (System Usability Scale) questionnaire, one of the best known and simplest methods of verifying the usability of a "system" [23]. The SUS questionnaire consists of 10 questions, and for each of them the user can answer on a scale of 1 to 5, where 1 means *Completely Disagree* and 5 means *Completely Agree*. The questions are as follows:

- 1) I think I would like to use this game often.
- 2) I find the game unnecessarily complex.
- 3) I found the game easy to use.
- 4) I think I would need help from a person with technical knowledge to use the game.
- 5) I think the various functions of the game are very well integrated.
- 6) I think the game has a lot of inconsistencies.
- 7) I imagine that people will learn how to use this game quickly.
- 8) I found the game awkward to use.
- 9) I felt confident using the game.

- 10) I needed to learn a lot of new things before I was able to use the game.

VII. RESULTS

As for comparison, the average SUS score is 68 [23]. In this work, the raw SUS score was 76.5. When converted into a percentile, the SUS score was 72.27, describing the game experience as "Good".

Regarding the qualitative testing with five participants, we focused on the paths participants took to achieve their goals in the game, what problems they had, and the comments they made. The results from the qualitative assessments revealed that some players clicked first the Biopsy button instead of the US & MRI one, even when the priority action was to ask for an ultrasound. This problem probably was due to the positioning of the Biopsy button on the Game home interface, which is located to the left of the US & MRI button, as shown in Fig. 3. The initiative to mitigate this problem when revising the game will be to invert the positions of these two buttons.

All the participants liked the visual game's aesthetics. They reported that the multiple management of clinical case scenarios was a very engaging experience, with comments reinforcing how excited and emotionally touched they were, respectively, during the winning (saving lives) and losing moments (losing lives).

The participants very much appreciated the sound effects and the musical theme. These audio elements helped to create reflective moments and to provoke decision-making actions demanding full player attention. They mentioned that ludic game aspects help to give them agency. In addition, in general, participants commented that the minimalist soundtrack brought concentration and a more engaging and immersive atmosphere.

Participants found representative the classic prostate cancer scenarios presented in the game. Four participants (three males and one female) expressed they had a meaningful interactive experience. Moreover, they commented about how important it is to focus on content related to the cancer condition to promote men's health.

Regarding the learning aspects of the game, participants who had family members and close friends with prostate cancer were able to make decisions more effectively. Perhaps this happens because they have some familiarity with the diagnostic process and/or with the medical tests commonly requested (we will need additional playtesting with more participants to clarify this issue). Two participants complained about the small font size of the patient input data displayed on the top panel of the Game home screen.

VIII. LIMITATIONS

Because the pandemic currently constrains human resources and time, this study was limited to generating online knowledge from the available participants. The study is also limited to a small sample of participants. It does not necessarily reflect other populations of users, such as people with no knowledge about prostate cancer or, on the contrary, people

highly specialized in this subject. Anyway, how players engaged in these prior experiences will indeed serve to enhance gameplay design. Future research efforts may need to include a representative number of Urologists, Oncologists, and Medical students. Efforts should involve these potential players to collect their feedback and use it for game revision and refinement to proceed from a prototype to a final executable game by polishing implementation details and adding custom game improvements over time.

IX. CONCLUSION AND FUTURE WORK

A game is a form of interactive entertainment where players must overcome challenges by taking actions governed by rules to meet a victory condition [24]. In the health area, the design of games that engage the audience and provide a meaningful experience is a complex task. However, when well designed, an interactive game can ease learning concepts and skills in the medical field, such as those related to prostate cancer, by managing multiple clinical case scenarios concurrently using decision-making. Overall, preliminary findings of players' experiences and perceptions of our game were very positive. During gameplay, the highly interactive nature and ludic game elements helped to give participants agency. Our findings also suggest that the gameplay's visual, textual, and audio elements may help increase the awareness and the generation of meaningful experiences for players. These experiences focus on the content of prostate cancer condition and on choosing the right medical actions in the proper order and time, without losing patients' lives.

Future work would benefit from using a data-driven approach to dynamically load many clinical cases of patients with prostate cancer symptoms based on a state machine that works on data containing their medical history and pre-existing conditions. Another necessary extension of this work is to create game levels with a purpose to amp up, even more, the player experience. In particular, we highlight the fundamental importance of heterogeneous team collaboration in this work, especially medical professionals. Finally, we will use the lessons learned with this preliminary usability study to refine the future versions of our prostate cancer game before being shared with a broader audience. We also plan to test the new revised versions of our game with specialized players in the short future.

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